Generation of neuronal variability and complexity

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The production of specialized differentiated neurons derived from stem cells has been proposed as a revolutionary technology for regenerative medicine. However, few examples of specific neuronal cell differentiation have been described so far. Although stem-cell tissue replacement might be seemingly straightforward in other cases, the high degree of complexity of the nervous system raises the challenge of tissue replacement substantially. Understanding mechanisms of neuronal diversification will not only be relevant for therapeutic purposes but might also shed light on the differences in cognitive abilities, personality traits and psychiatric conditions observed in humans.

The complexity of the human brain permits the development of sophisticated behavioural repertoires, such as language, tool use, self-awareness, symbolic thought, cultural learning and consciousness[1]. Partly owing to the uniqueness of the neuronal heterogeneity and interconnections in our brains, each human is different. Indeed, the ability to perform high-level cognitive functions has been shown to be directly influenced by individual memory capacity, brain complexity and neural activity[2]. Such complexity is the result of millions of years of evolution that have equipped neural progenitor cells in the embryo with the ability to generate every neuron in the brain — an extraordinary example of cellular potency. Other mammals, such as mice and chimpanzees, also have high levels of cellular heterogeneity, but the level of complexity in their brains is believed to be lower than that observed in humans. The mouse brain contains approximately 75 million neurons[3]. By contrast, the human brain is composed of approximately 100 billion neurons. The total number of synapses in the human neocortex is estimated to be 0.15 quadrillion (10^{15}), and a typical neuron has approximately 5,000–200,000 synapses[4]. The diversity of single neurons provides the underpinning for how neuronal circuits operate; therefore, it is essential to identify the participating neuronal subpopulations, and to determine the functions of the neurons at the cellular and molecular levels.

An extraordinary diversity is found within the neuronal population. There might be as many as 10,000 types of neuron, although the definition of neuronal type is debated. Diversity exists between individual cells within a neuronal subtype, highlighting the importance of single neurons in the network. Even neurons with a similar morphology can differ in important molecular details, expressing different combinations of ion channels, for instance, providing cells with various excitation thresholds and distinctive firing patterns. However, how and when neuronal diversity is generated remains unknown. Here we review several molecular mechanisms that contribute to neuronal diversity.

Origins of neuronal diversity

Neurons of the mammalian central nervous system (CNS) arise during embryonic development from neuroproliferative zones surrounding the ventricles of the neural tube[5]. For a defined period of time, all types of postmitotic neuron must be formed from precursor neuroblasts that exit the cell cycle and migrate from these regions to form the adult grey matter. During embryonic neurogenesis there is a massive proliferation of cells, but only 15–40% of post-migratory cells survive[6,7], suggesting that a selection process is occurring[8]. Interestingly, neurogenesis continues in the adult brain in discrete areas, such as the anterior part of the subventricular zone and the subgranular zone of the dentate gyrus (DG) of the hippocampus[9–11].

DNA rearrangement was first suggested to be responsible for the generation of neuronal diversity in a fashion similar to that seen in the somatic DNA rearrangement of the immune system, which is known as the V(D)J mechanism[12]. In the immune system, the highly diverse array of antigen receptors can be attributed to the stochastic nature of the recombination process in somatic precursor cells, causing permanent changes in DNA and gene expression. This diverse population is then the target of selective processes that favour the correct antigen–receptor match and eliminate those with inadequate specificities, accounting for the rapid kinetics and immense diversity observed *in vivo*. In the nervous system, diversity is high during initial development[13] and increases owing to activity in a more mature brain (Fig. 1). The activity-dependent neuronal diversity in postmitotic neurons is driven by environmental stimuli that dynamically refine neuronal networks, such as in the hippocampal 'place cells'[14]. Yet how the original diversity between neurons is generated remains to be uncovered. Evidence for a possible similarity between the nervous and immune systems came from studies on mice deficient in DNA double-strand-break repair, revealing impaired neurogenesis as well as lymphogenesis, suggesting that normal neuronal development requires essential components of the conserved non-homologous end-joining pathway[15]. However, attempts to demonstrate a programmed recombination event in the CNS have led to unreconciled controversy[16,17]. Rigorous but not definitive arguments against this hypothesis have come from experiments that used the genome of a single olfactory receptor neuron to clone an entire mouse (see page 1061); no evidence for recombination was found[18,19].

In the next sections, we review some of the candidate molecular mechanisms that can actively contribute to the generation of neuronal diversity, acting on DNA, RNA and protein substrates.

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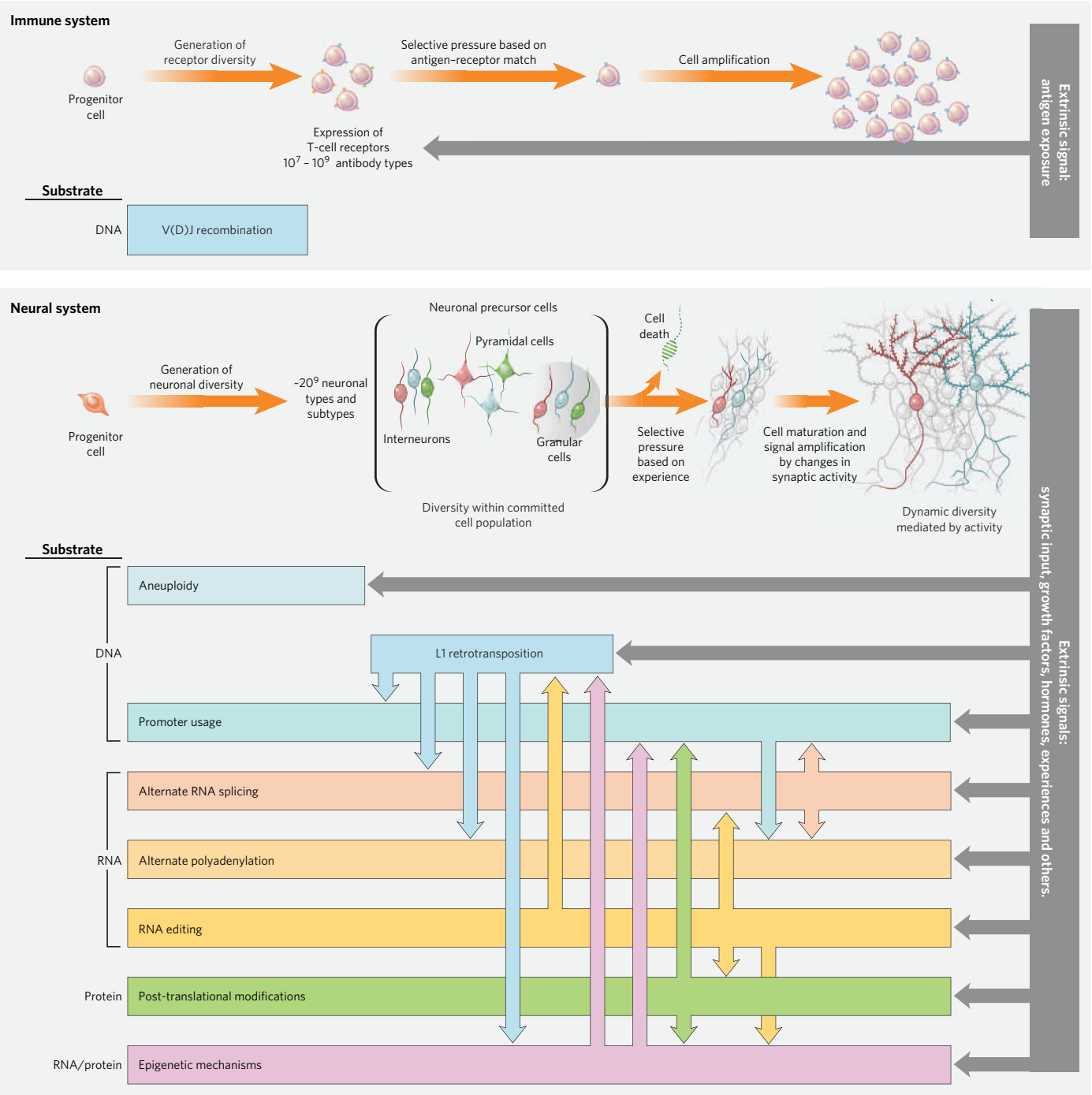

Figure 1 | Generation of somatic variability in the immune and neural systems. In the immune system, vast numbers of antibodies are generated by the V(D)J recombination system. In the nervous system, neuronal diversity is twice as high as the cellular diversity produced by the immune system. There is no single intrinsic mechanism that can explain such diversity, but rather a combination of several systems acting in different substrates (DNA, RNA or proteins) might be involved in neuronal differentiation. Finally, a dynamic diversity is generated by experience, correctly tuning neuronal wiring and generating diversity within subtypes of neuron. Contrary to the immune system, extrinsic factors can act directly on the intrinsic mechanisms of neuronal diversity and not only on selected cells. Moreover, intrinsic events that occur early during differentiation might only contribute to diversity in a more mature brain.

Aneuploidy

Aneuploidy is the loss or gain of chromosomes to produce a numerical deviation from multiples of the haploid chromosomal complement. Such chromosomal alteration has obvious consequences on cellular physiology, leading to well-documented diseases, such as Down's syndrome. Occasional somatic aneuploidy has frequently been associated with cell death[20] or cancer[21]. The original observation that aneuploid neural progenitor cells can divide and differentiate *in vitro* was attributed to a clonal tissue-culture artefact[22]. Chromosomal variation in neurons and glia of the developing and adult mammalian nervous system was mainly detected by fluorescence *in situ* hybridization (FISH) in brain tissues of rodents and humans[23,24]. From studies in mice, it is estimated that approximately 33% of neural progenitor cells display chromosomal aneuploidy that encompasses both loss and gain of chromosomes. However, the FISH technique presents potential artefactual problems that might interfere with sample analysis[25]. Moreover, the overall percentage and complete chromosomal complement of aneuploid cells, hypoploidy, hyperploidy and several chromosomal combinations remain to be determined. Mature aneuploid neurons were described as being functionally active and integrated into brain circuitry, suggesting that a

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network composed of intermixed euploid and aneuploid neurons might produce unique signalling properties, owing to its exceptional genetic-expression profile[26]. Even the quantitative change of a single gene can produce major changes in cellular signalling[27], so the effects associated with loss or gain of approximately 1,000 gene copies per chromosome should produce even more drastic alterations in a cell's physiology and, consequently, in the neuronal network. However, the role of non-neuronal aneuploid (that is, glial) cell types is less obvious. No tissue-specific factor has been described that contributes to aneuploidy in the nervous system, and it seems that these cells arise, in part, via a chromosome mis-segregation mechanism during mitosis[28] (Fig. 2).

LINE-1 retrotransposition

The finding that LINE-1 retrotransposons (L1s) can have an effect on neuronal genomes challenges the dogma that neurons are genetically homogeneous units[29]. L1s are among the most ancient and successful elements in eukaryotic genomes, comprising approximately 20% of mammalian genomes[30–32]. Active human L1s are about 6 kb long, harbouring an internal promoter and encoding two open reading frames (ORFs) — an RNA chaperone protein, ORF1, and an endonuclease and reverse transcriptase (RT) protein, ORF2 — usually ending with a short tail that is rich in adenine bases. Upon translation, the L1 RNA assembles with its own encoded proteins[33] and moves back to the nucleus, where the endonuclease makes a single-stranded nick and the RT uses the nicked DNA to prime reverse transcription from the 3´ end of the L1 RNA. Moreover, the L1-encoded proteins can function *in trans* to mobilize non-autonomous retrotransposons (such as Alu elements) and cellular messenger RNAs, generating processed pseudogenes[33–36]. L1s can threaten the structure and regulate the expression of the genome in different ways, such as insertional mutagenesis[37,38]. L1s are probably silenced in neural stem cells due to SOX2-mediated transcription repression[29]. Downregulation of *Sox2* accompanies chromatin modifications, such as DNA demethylation and histone acetylation, which, in turn, might trigger neuronal differentiation (Fig. 3). Despite the low number of L1 insertions characterized, the sequence data from target insertional sites in rat neuroblasts were often close to, or inside, neuron-associated genes, indicating that the integration process might be regulated[29]. Some of these target genes included those for an olfactory receptor, ion-channel-associated genes and a cadherin receptor[29]. It is challenging to estimate precisely the potential activity of endogenous L1s in neuroblasts. Most RNA molecules can be subject to retrotransposition if hijacked by L1 machinery[33–36]. Thus, each developing neuron can potentially carry L1-mediated events and, if part of the resultant insertions occurs in genes expressed during neuronal development, it is possible that brain development could be significantly affected by L1s.

Promoter usage and multiple start sites

Promoters and enhancers are *cis*-acting elements that control gene transcription via a complex arrangement of protein–DNA and protein–protein interactions[39]. For decades, the concept that these regulatory mechanisms control only the quantity, and not the identity, of their corresponding transcripts dominated our conception of gene expression. Many eukaryotic genes contain multiple promoters, each subjected to different regulatory factors present in a cell at a certain moment[40]. By definition, each promoter determines a different start site and first exon, and, consequently, a different transcript. Frequently, transcripts arising from genes with multiple promoters only differ in their 5´ non-coding regions, sharing the same ORFs (Fig. 4). Minimal promoter sequences or enhancers can be used to selectively drive transcripts in a spatially and temporally specific manner[39]. Multiple transcriptional start sites are a common attribute of the glutamate receptor ion-channel genes. The expression of the various glutamate receptor genes is under developmental control and often responds to environmental stimuli. In several cases, the promoter regions include overlapping motifs for the transcription factors Sp1 and GSG near the major initiation sites, and a silencer element to guide expression in neurons[41] (Fig. 4).

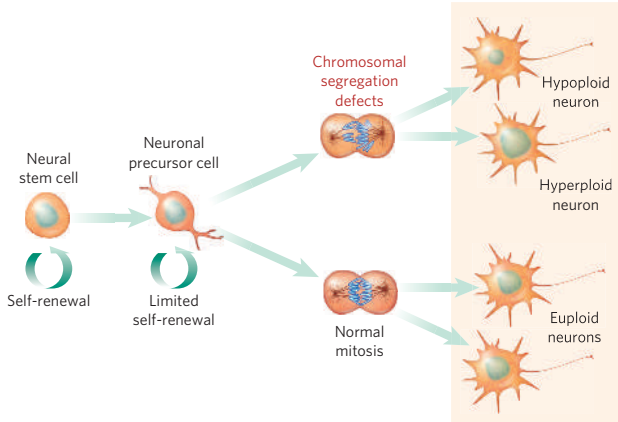

Figure 2 | Aneuploidy as a mechanism to alter gene expression in the CNS. **Chromosomal segregation defects during mitosis can generate aneuploid neurons and astrocytes[28], resulting in a euploid population intermixed with an aneuploid population of cells.**

A better understanding of the regulation of genes with multiple start sites will require the identification of the conditions under which each promoter becomes activated or repressed. The brain-derived neurotrophic factor (*BDNF*) gene contains a complex pattern of regulation, including the use of a variety of promoters. Four promoters have been mapped upstream of the 5´ exons; these are differentially controlled by the Ca^{2+} signals evoked via *N*-methyl-D-aspartate (NMDA) glutamate receptors or L-type voltage-dependent calcium channels (L-VDCCs)[42]. The diversity of regulatory sequences could allow different constellations of transcription factors that are present in discrete parts of the brain to regulate BDNF in ways that produce differential patterns of isoform expression. Studies on the effect of acute cocaine exposure reveal promoter-specific upregulation of specific BDNF transcripts in the striatum and frontal cortex, whereas exposure to kainic acid can stimulate (by up to 50-fold) other BDNF transcripts in the hippocampus, implying the existence of different forms of synaptic plasticity in the two systems[43,44].

Alternative RNA splicing

Precursor RNAs made by eukaryotic RNA polymerase II (pol II) undergo a series of post-transcriptional processing events to produce mature mRNAs, which are then exported to the cytoplasm. Physiological alterations in RNA processing can serve as important control points of gene expression in a tissue or during development[45]. Alternative RNA splicing is a versatile mechanism for generating a large repertoire of functionally unique proteins and occurs particularly frequently in the nervous system[46] within neuronal genes, such as those for neurotransmitter receptors, ion channels, synaptic surface receptors, axon-guidance receptors and transcription factors[47]. The importance of alternative splicing in the nervous system became evident following the association of aberrantly spliced mRNAs with several neurological diseases[48]. A striking example of complex alternative splicing is the *Drosophila* axon-guidance receptor (DSCAM), which is a homologue of the human Down's syndrome cell-adhesion molecule. The alternatively spliced exons each encode immunoglobulin half-domains that combine to create a receptor with as many as 38,016 different protein variants — double the predicted number of genes in the *Drosophila* genome[49]. The mammalian homologue of *Dscam* is not subject to similar levels of alternative splicing; however, other mammalian genes implicated in cell-surface recognition and synapse formation, such as the neurexins, can generate numerous isoforms[50]. Cadherin-related neural receptors (CNRs or protocadherins) are encoded in an arrangement that is intriguingly reminiscent of the antibody genes, with 52 variable exons in humans, each encoding an entire extracellular and transmembrane region, grouped in three tandem arrays upstream of a single set of constant exons encoding the cytoplasmic region[51].

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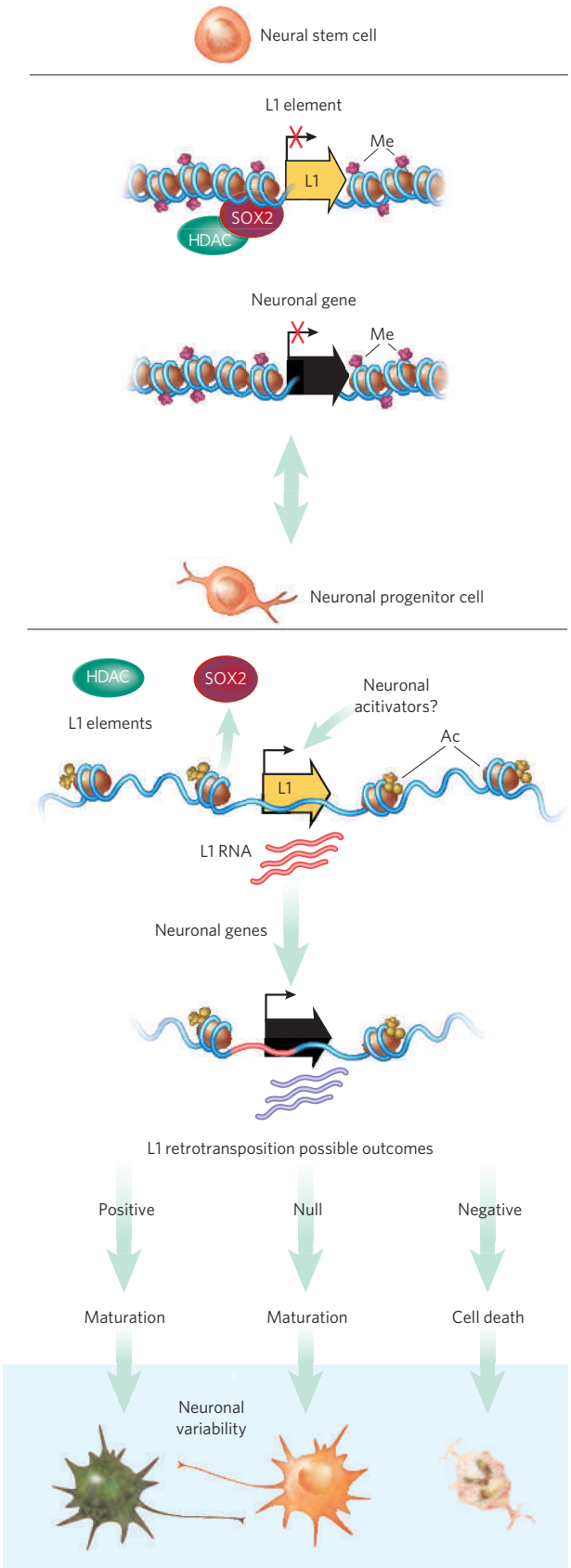

Figure 3 | A model for generation of neuronal diversity by L1 retrotransposition. During neuronal differentiation, L1 transcription is activated by chromatin modifications and release of SOX2, allowing subsequent retrotransposition into neuronal genes[29]. The resulting retrotransposition events can alter gene expression, which can, in turn, influence the phenotype of the resulting cell. HDAC, histone deacetylase.

Promoter choice and splice-site selection seem to be responsible for the expression of individual protocadherin mRNAs[52]. Alternative splicing commonly regulates molecules that mediate cell excitation. Changes in the splicing of these transcripts in response to electrical activity represent an attractive additional mechanism for inducing long-term changes in excitability. Indeed, splicing of NMDA receptor 1 transcripts can be altered in response to various stimuli[53,54].

Differences in alternative splicing from individual cells in the brain have been examined using single-cell reverse transcription (RT)–polymerase chain reaction (PCR) combined with patch-clamp recordings[55]. These experiments revealed that differences in splicing from cell to cell are apparent even within a defined population of neurons. Moreover, neurons in different regions of the brain display splicing differences that are consistent with their function, suggesting a sophisticated regulatory mechanism[55]. Mechanisms for alternative splicing control in neurons can include cell excitation, splice-site choice, RNA-binding proteins, splicing enhancement, exon definition and neuron-specific RNA factors, such as Nova, Hu and neural polypirimidine tract-binding proteins[48,56]. The recent demonstration that the methyl-CpG-binding protein 2 (MeCP2), which has been implicated in Rett syndrome, can act as a splicing factor, linking transcription splicing to epigenetics, indicates that other factors are likely to exist[57].

Alternative polyadenylation

mRNAs synthesized in the nucleus generally contain a 3′ poly(A) tail; the rare exceptions are the transcripts of replication-dependent histone genes. A functional polyadenylation signal is required for transcription termination by RNA pol II and transport from nucleus to cytoplasm, where the poly(A) tail on the message stabilizes the mRNA against degradation. Poly(A) sites can directly influence the amount of cytoplasmic RNA produced from a transcript; therefore, alterations in polyadenylation efficiency can have a profound effect on the amount and nature of a gene product. Alternative polyadenylation can be a direct consequence of tandem poly(A) signals within a single 3′ untranslated region (UTR), of competition between promoter-proximal polyadenylation and splicing sites, or of the presence of alternative 3′-terminal exons that can be skipped in some circumstances. All of these strategies can produce proteins differing at their carboxyl termini, depending on which processing pathway is followed (Fig. 5). Many genes have been characterized as having an alternative polyadenylation site at the 3′ end of their mRNAs, which can be preferentially used depending upon the cellular environment[58]. In the nervous system, several examples of alternative polyadenylation have been described. *MeCP2* encodes four different transcripts distinguished by the length of the 3′ UTR. The long (10 kb) mRNA form is highly expressed in a subpopulation of cells in the adult brain, in contrast to the shorter transcripts found in the lungs and liver[59]. Moreover, the developmental dynamic of MeCP2 in the brain suggests a potential role of the 3′ UTR in the regulation of protein synthesis in response to age-specific requirements of MeCP2 function[60].

Alternative polyadenylation seems to be more abundant in the nervous system than in other tissues[61]. The study of pre-mRNA cleavage and polyadenylation has revealed several factors required for accurate RNA processing, including cleavage and polyadenylation-specificity factor (CPSF), cleavage-stimulatory factor (CstF), poly(A) polymerase (PAP), the cleavage factors Im and IIm, and the poly(A)-binding proteins (PABPs). These proteins interact to form a complex on the precursor RNA prior to the cleavage and polyadenylation reactions[62]. Determination of the differential expression levels of these factors in the nervous system throughout development will provide a better understanding of the differential use of multiple poly(A) sites in complex transcription units.

RNA editing

Similar to RNA splicing, RNA editing alters the sequence of RNA from the encoded DNA by changing as few as one or two nucleotides. Alterations include converting one base to another, removing one nucleotide and substituting another, deleting encoded residues and inserting

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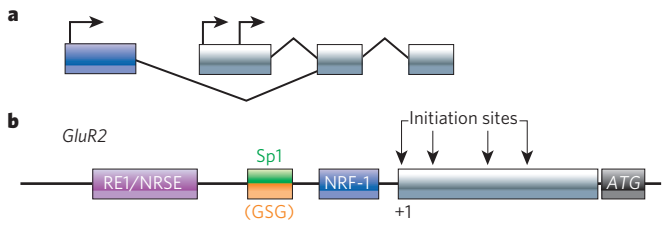

Figure 4 | Alternative start sites. a, Alternative promoter regions, 5´-splice sites or start sites can alter the length of the mRNA. **b,** Schematic of the proximal promoter region for the *GluR2* gene. The neuron-restrictive silencer factor REST/NRSF can repress the GluR2 promoter via the neuron-restrictive silencer element RE1/NRSE. Apart from the silencer, Sp1 and nuclear respiratory factor-1 (NRF-1), regulatory elements within 60 base pairs of the +1 initiation site are required for strong promoter activity in cortical neurons. A potential GSG motif is also present, overlapping with the Sp1 binding site.

non-template nucleotides. Two types of RNA editing have been found in nuclear-encoded mRNAs, both of which involve deamination of encoded nucleotides: the first involves the deamination of cytidine (C) to create uridine (U), and the second involves the deamination of adenosine (A) to create inosine (I). Both types of editing can cause changes at the amino-acid level and frame shifts in mRNAs, creating *de novo* ORFs. They can also affect RNA secondary structure, such as in transfer RNAs and ribosomal RNAs, and can modify start and stop codons. Other processes that can be affected include RNA splicing, transport and stability; however, not all editing events have obvious effects, as some codon changes are silent or fall within non-coding regions of mRNA. Editing can be regulated in a developmental, environmental, hormonal or tissue-specific manner[63], and is most abundant in mRNAs in the nervous system[64]. The mammalian editing enzymes, adenosine deaminases acting on RNA (ADARs), are widely expressed in several tissues; however, all known edited RNAs occur only in the CNS. The importance of such enzymes was revealed through gene knockouts in different systems. Although ADAR knockouts in flies and worms lead to behavioural abnormalities[65,66], knockouts in mammals have shown them to be essential for life[67].

Several subunits of glutamate receptor channels contain a glutamine residue that is subjected to A-to-I editing of the CAG codon into an arginine residue (Q to R in the single-letter amino-acid code) that changes a key residue within the ion pore of the channel. The unedited Q form of the GluR-B subunit renders α-amino-3-hydroxy-5-methyl-4-isoxazole propionic-acid (AMPA) receptors more permeable to calcium ions. This substitution can have several functional consequences, such as voltage-independent gating, decreased single-channel conductance and permeability to divalent cations, as the new amino acid affects the hairpin loop that forms the narrow constriction of the channel[68]. Specific Q/R site editing in the GluR6 transcript was demonstrated to be relevant for receptor-independent long-term potentiation at a point at which the medial perforant path synapses on the DG granule cells, increasing the vulnerability to kainite-induced seizures[69]. These findings indicate a clear role for RNA editing of GluR6 in synaptic plasticity, circuit excitability and seizure susceptibility. Furthermore, mRNA glutamate receptor editing is crucial for neuronal survival, and reduction of RNA editing might be a direct cause of selective motor-neuron death in amyotrophic lateral sclerosis patients[70]. The serotonin receptor gene can generate 18 different complementary DNA (cDNA) sequences and 12 different predicted protein isoforms, owing to partial editing at the five A-to-I sites in a region involved in coupling to G proteins[71]. Moreover, some edited isoforms are differently expressed in distinct brain regions and have different affinities for some antipsychotic drugs, suggesting that predicted protein products are likely to be functionally important[71,72]. Thus, the levels of editing in this receptor might be influenced by factors that alter levels of serotonin in the brain. Altered editing levels have been observed in schizophrenic patients, and might affect patients with Huntington's disease and Alzheimer's disease[73]. Depressed suicide victims were also reported to have altered transcript-editing levels in the prefrontal cortex[74].

Post-translational modifications

Post-translational modifications of cellular proteins can regulate protein stability, activity and localization. Protein modification by covalent attachment of radicals to target proteins, such as ubiquitination, sumoylation or phosphorylation, is rapid and reversible, being a potent mechanism for the spatiotemporal control of protein activity, especially at the synapse level[75]. The beauty of protein-modification choreography can be appreciated in the control of synaptic differentiation by the transcription myocyte enhancer factor 2 (MEF2). Phosphorylation of a serine residue in MEF2 prevents activation of its target genes, whereas dephosphorylation can be induced by calcium influx, triggering MEF2-dependent transcription in hippocampal neurons. In cerebellar granule neurons, the phosphorylated isoform of MEF2A can switch from activator to repressor by sumoylation, suggesting that phosphorylated and sumoylated MEF2A represses transcription of target genes, promoting synaptic differentiation. This sumoylation switch might underlie the differential effect of MEF2 on synapse number in cerebellar and hippocampal neurons[76,77]. Post-translational regulation can also induce epigenetic modifications through chromatin modification, regulating expression in a set of genes (see below).

Epigenetic mechanisms

Epigenetic modifications, including DNA methylation and histone modification (such as acetylation and methylation), can drive a specific, reversible and heritable pattern of gene expression on differentiating cells, without affecting the DNA sequence. Such mechanisms allow changes in chromatin structure, coordinating the activation/repression of gene arrays at precise times during development. Although several examples of neuronal plasticity exist, the study of epigenetic modifications in cellular diversity is in its infancy[78,79]. The combinatorial methylation of specific lysine residues at the amino-terminal tails of the histones H3 and H4 is increasingly accepted as a molecular imprint involved in the regulation of gene expression in eukaryotes[80]. Interestingly, levels of dimethylated and trimethylated histone H3 at lysine 4 showed significant differences between human glutamate promoters in different brain regions and during development, reinforcing the idea of histone methylation as a transcriptional regulator in the CNS[81]. DNA methylation is another type of epigenetic modification that mediates gene silencing by inhibiting transcription-factor binding to promoter regions or by binding methyl-CpG-binding proteins, such as MeCP2, leading to transcriptional repression. One of the well-known functions of DNA methylation is to suppress transcriptional activity of endogenous retro-elements, including L1s. There is evidence that endogenous retroviral transcripts are expressed in the nervous system of patients with schizophrenia and other complex psychotic diseases; however, the role of all these transcripts is obscure[82–87]. Theoretically, such an abundance of endogenous transcripts could represent precursor RNAs to several

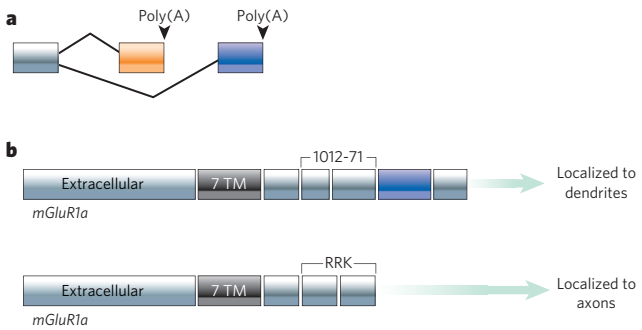

Figure 5 | Alternative poly(A) sites. a, Choosing from several distinct polyadenylation sites, transcripts with 3´-differential exons can be generated. **b,** Truncation of the GluR1 carboxy-terminal by alternative splicing redirects the protein product from neuronal dendrites (dominant signal 1012-71) to axons (RRK motif)[96]. 7 TM, seven transmembrane-spanning region.

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non-coding small RNAs with potential regulatory functions in the cell. Alternatively, these endogenous retroviruses might be more active in the brains of affected individuals and contribute to the phenotype.

Non-protein-coding RNAs can also cause post-transcriptional modification and might influence epigenetic modifications. They seem to be mostly expressed in the brain, suggesting that individual gene expression or clusters of genes in the nervous system can be highly affected by the action of these versatile molecules[88,89]. MicroRNAs (21–25 nucleotides) are derived from longer RNA precursors, processed in the nucleus and subsequently exported to the cytoplasm, where they associate with the RNA-induced silencing complex (RISC) that directs the microRNAs to their targets for post-transcriptional repression[90]. Computational algorithms have suggested that microRNAs have the ability to simultaneously regulate many targets, providing a means of coordinated control for concerted gene action[91]. Despite the scarcity of functional examples, data regarding predicted targets, expression patterns and overexpression studies suggest that microRNAs are attractive key players for differential gene expression in many neuronal types.

Concluding remarks

Understanding what produces neuronal diversification has been a long-standing challenge for neuroscientists. More than a century ago, Camillo Golgi published a technique for the silver staining of isolated neurons in samples of highly dense cells in a brain slice. The Golgi technique was important because it allowed Ramon y Cajal to realize the incredible morphological diversity of neurons, directly reflecting their synaptic connections. Nowadays, the production of such neuronal diversity from a relatively homogeneous population of neural stem cells in culture has great promise for the treatment of several neurological diseases (see page 1094).

The evolution of several genetic strategies in eukaryotes greatly expanded the number of functionally distinct proteins that could be produced from a given gene pool. However, are all these strategies required or even adequate to generate such complex neural networks? The answer is probably no. Almost certainly, other mechanisms remain to be uncovered and understood. Emerging studies of non-coding RNAs have provided us with tantalizing examples of regulatory molecules[88]. For instance, small double-stranded modulatory RNAs have been proposed to regulate the generation of neurons from adult neural stem cells by working as a transcriptional co-activator for a large set of neuronal genes[92]. The combined use of several systems, along with new neuron-specific mechanisms that provide extra adaptability to an individual process (for instance, L1 insertions creating new splicing sites or polyadenylation signals[93–95]) and extrinsic signals, is likely to be responsible for generating the large spectrum of protein activities that supports complex cognitive brain functions, such as consciousness and plasticity. ■

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